

KWESI A. QUAGRAINE, Ph.D.

Chief Scientist, Climate Vulnerable Forum and Vulnerable Twenty Group (CVF-V20) Secretariat

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PROFILE

Climate scientist and science diplomacy practitioner bridging Earth system research, multilateral engagement and climate policy for vulnerable countries. As Chief Scientist of the CVF-V20 I lead the science and health agenda for a group of climate vulnerable economies representing 1.7 billion people, translating complex scientific evidence into policy-relevant analysis for ministers, principals and multilateral processes. Publications in Nature and Nature Climate Change on climate governance, with a record of advisory work for WMO, UNEP and the Green Climate Fund.

EXPERIENCE

Chief Scientist | CVF-V20 Secretariat, Accra, Ghana

2026 – present

- Lead the science and health agenda for the 1.7 billion people of the climate vulnerable member economies.
- Issue a monthly versioned climate risk brief series to member states as the Office of the Chief Scientist, tracking ENSO developments and regional outlooks.
- Prepare briefing documents for principals engaging ministerial and regional processes, including CARICOM engagement on health-related climate loss and damage.
- Develop funder-facing science programs with philanthropic partners on ocean-based carbon dioxide removal, corals and climate-health workstreams.

Postdoctoral Fellow, then Visiting Scientist | NCAR, Boulder, CO

2023 – present

National Center for Atmospheric Research

- Led research on stratospheric aerosol injection (SAI) impacts on African precipitation extremes using CESM, including regionally refined variable-resolution modeling.
- Developed novel indices for subtropical high-pressure systems central to African climate variability.

Senior Lecturer | Department of Physics, University of Cape Coast, Ghana

2023 – present

- Lecturer 2020 – 2023. Lead academic programs in atmospheric science and meteorology for 100+ students annually and mentor graduate researchers from climate vulnerable regions.

Research Fellow | Climate System Analysis Group, University of Cape Town, South Africa

2016 – 2023

- FRACTAL project (UK DFID). Worked directly with city decision-makers across Southern Africa to integrate regional climate understanding into urban planning and climate-sensitive decisions.

POLICY ADVISORY AND CONSULTANCIES

- UNEP and Green Climate Fund. Ghana National Adaptation Plan, enhancing multi-sector adaptation planning and capacity for national decision-makers.
- World Meteorological Organization. Developed a User Interface Platform framework for operationalizing climate services worldwide.
- ICLEI Local Governments for Sustainability. Regional climate projections for urban planning in Bo City, Sierra Leone and Cape Coast, Ghana (2021 – 2022).
- IPCC Expert Reviewer for assessment report review processes.

EDUCATION

Ph.D., Climatology | University of Cape Town, South Africa

2021

M.Phil., Physics (Meteorology and Atmospheric Physics) | University of Cape Coast, Ghana

2014

B.Sc., Physics | University of Cape Coast, Ghana

2011

RESEARCH LEADERSHIP AND FUNDING

- Co-Investigator, DEGREES SRM Governance Initiative, Modeling Research Fund (2023 – 2027): SAI interactions with the Harmattan and climate impacts over Southern West Africa.
- Co-Investigator, DEGREES SRM Governance Initiative, Socio-Political Research Fund (2024 – 2027): SRM knowledge and governance across Ghana.

- Co-Investigator, Reflective High Latitude Low Altitude (HiLLA) Research Grant (2025 – 2026): HiLLA impacts on the West African Monsoon.
- NRF SARCHI Scholarship in Climate Change, South Africa (2016 – 2023); Newton Research Fellow, Climatic Research Unit, University of East Anglia, UK (2017); NCAR CISL supercomputing allocation recipient.

SELECTED PUBLICATIONS (25+ PEER REVIEWED)

1. **Quagraine, K. A.**, Lynas, M. and Ellis, E. C. (2026). Why we must replace temperature limits with clean energy targets. *Nature*.
2. **Quagraine, K. A.**, Abiodun, B. J. and Essien-Baidoo, S. (2025). Africa must lead the governance of solar radiation management. *Nature Climate Change*.
3. **Quagraine, K. A.**, Tye, M. R., Quagraine, K. T., Tilmes, S., Simpson, I. R. et al. (2025). Impact of stratospheric aerosol injection on precipitation extremes across Africa. *Environmental Research: Climate*.
4. Richter, J. H., Joseph, E., Arcodia, M. C., ..., **Quagraine, K. A.** et al. (2026). Earth system predictability across time scales for a resilient society. *Bulletin of the American Meteorological Society*.
5. **Quagraine, K. A.**, Hewitson, B., Jack, C. et al. (2020). Using co-behavior analysis to interrogate the performance of CMIP5 GCMs over Southern Africa. *Journal of Climate*, 33(7), 2891-2905.
6. Nkrumah, F., **Quagraine, K. A.**, Quenum, G. M. L. D. et al. (2025). Assessing regional climate trends in West Africa under geoengineering. *Journal of Geophysical Research: Atmospheres*.

Full list: <https://scholar.google.com> (*Kwesi A. Quagraine*)

CAPACITY BUILDING AND COMMUNITY LEADERSHIP

- Short courses on solar radiation management and carbon dioxide removal at three Ghanaian universities (2024); invited talks including AGU Fall Meetings (2018, 2020, 2022, 2024).
- Recycle Up! Ghana with Technik Ohne Grenzen (2016 – present): recycling education reaching 3,000+ students, 6 summer camps and 10 school recycling groups.
- Peer reviewer for *Nature Geoscience*, *Journal of Climate*, *Environmental Research: Climate* and other journals. Languages: English (fluent), Fante (native).